ASSIGNMENT-4

NLTK (Natural Language Toolkit)

It is a Python library used for processing and analyzing human language data in NLP applications.

NLTK (Natural Language Toolkit) is an open-source Python library used for:

- Text processing
- Tokenization
- Lemmatization
- Stemming
- POS tagging
- Sentiment analysis
- Building basic chatbots

It is widely used for **learning and implementing NLP concepts**.

NLTK is Important in Chatbots

In traditional chatbots (before LLMs), NLTK helps:

- Break sentences into words (Tokenization)
- Convert words to base form (Lemmatization)

Example Using NLTK

User Input:

"I am feeling stressed."

Using NLTK:

- Tokens → ['I', 'am', 'feeling', 'stressed']
- Lemma → feel, stress
- POS → Pronoun, Verb, Verb, Adjective

This helps the chatbot understand the user's emotion.

How to Install NLTK (VS Code)

In terminal:

```
pip install nltk
```

Then download required resources

```
import nltk
nltk.download('punkt')
nltk.download('wordnet')
nltk.download('averaged_perceptron_tagger')
```

NLTK in NLP Pipeline

Flow:
User Input → Tokenization → Lemmatization → POS Tagging → Intent Detection → Response

User Input

Raw text entered by the user.

"I am feeling stressed and anxious."

This is unstructured natural language.

Tokenization

Breaking sentence into smaller units (words).

Output:

['I', 'am', 'feeling', 'stressed', 'and', 'anxious']

👉 Helps system analyze each word separately.

Welcome | download_nltk.py | tokenization.py

tokenization.py

```python
import nltk
from nltk.tokenize import word_tokenize

# Sample sentence
text = "I am feeling stressed and anxious."

# Tokenization
tokens = word_tokenize(text)

print("Original Text:")
print(text)

print("\nTokens:")
print(tokens)
```

```
(venv) C:\Users\geeth\OneDrive\Documents\Assignment-4>python tokenization.py
Original Text:
I am feeling stressed and anxious.

Tokens:
['I', 'am', 'feeling', 'stressed', 'and', 'anxious', '.']
```

Lemmatization

Convert words to base form.

Output:

['I', 'be', 'feel', 'stress', 'and', 'anxious']

Examples:

- feeling → feel
- stressed → stress

Welcome | download_nltk.py | tokenization.py | lemmatization.py

lemmatization.py

```python
import nltk
from nltk.stem import WordNetLemmatizer
from nltk.tokenize import word_tokenize

text = "The patients were running and feeling better."

lemmatizer = WordNetLemmatizer()
tokens = word_tokenize(text)

lemmas = [lemmatizer.lemmatize(token.lower()) for token in tokens]

print("Original Words:")
print(tokens)

print("\nLemmatized Words:")
print(lemmas)
```

```
(venv) C:\Users\geeth\OneDrive\Documents\Assignment-4>python tokenization.py
Original Text:
I am feeling stressed and anxious.

Tokens:
['I', 'am', 'feeling', 'stressed', 'and', 'anxious', '.']

(venv) C:\Users\geeth\OneDrive\Documents\Assignment-4>python lemmatization.py
Original Words:
['The', 'patients', 'were', 'running', 'and', 'feeling', 'better', '.']

Lemmatized Words:
['the', 'patient', 'were', 'running', 'and', 'feeling', 'better', '.']
```

POS Tagging

Identify grammatical role of each word.

Output:

I → Pronoun
am → Verb
feeling → Verb
stressed → Adjective
anxious → Adjective

This helps system understand:

- User emotion = stressed, anxious
- Action = feeling

```python
import nltk
from nltk.tokenize import word_tokenize
from nltk import pos_tag

text = "The patient is feeling anxious."

tokens = word_tokenize(text)
pos_tags = pos_tag(tokens)

print("Tokens:")
print(tokens)

print("\nPOS Tags:")
print(pos_tags)
```

```
(venv) C:\Users\geeth\OneDrive\Documents\Assignment-4>python pos_tagging.py
Tokens:
['The', 'patient', 'is', 'feeling', 'anxious', '.']

POS Tags:
[('The', 'DT'), ('patient', 'NN'), ('is', 'VBZ'), ('feeling', 'VBG'), ('anxious', 'JJ'), ('.', '.')]
```

Intent Detection

System identifies what user wants.

Example intents:

- Greeting
- Symptom query
- Mental health support

For our example:

Intent = Mental Health Support

```python
text = input("Enter your message: ")

if "stress" in text.lower() or "anxious" in text.lower():
    intent = "Mental Health Support"
elif "fever" in text.lower():
    intent = "Physical Health Query"
else:
    intent = "General Query"

print("Detected Intent:", intent)
```

```
(venv) C:\Users\geeth\OneDrive\Documents\Assignment-4>python intent_detection.py
Enter your message: I feel anxious
Detected Intent: Mental Health Support
```

Response Generation

Now chatbot generates reply:

“I’m sorry you’re feeling stressed. Try deep breathing exercises and consider taking short breaks.”

In LLM-based systems like those from OpenAI, this step is handled by the language model automatically.